Charge, Mass, and Inertia as Saturated Responses in a Pre-Geometric Relational Description

Jérôme Beau[1*]

[1*]Independent Researcher, France.

Corresponding author(s). E-mail(s): jerome.beau@cosmochrony.org;

Abstract

We investigate the structural origin of mass, electric charge, and inertial response within a pre-geometric relational framework in which physical observables arise through a generally non-injective projection onto effective spacetime descriptions. In this setting, geometric and dynamical quantities are not fundamental but emerge as regime-dependent responses of an underlying relational description subject to intrinsic saturation bounds.

We show that mass and charge can be consistently interpreted as distinct symmetry realizations of a single bounded-response mechanism. Mass corresponds to an isotropic inhibition of relational relaxation, while electric charge arises as an oriented saturation of the same underlying flux. Inertia emerges as the linear-response limit of a bounded projective update process, rather than as an intrinsic resistance to motion.

In addition to the conceptual analysis, we provide an explicit effective realization of the saturation mechanism in Lagrangian form. A finite projective capacity necessarily induces Born–Infeld–type nonlinearities, from which mass-like spectral gaps, bounded inertial response, and local $U(1)$ charge structure arise as symmetry consequences of saturation, corresponding to the phase fiber associated with non-injective projection of the underlying relational configuration space.

The proposed framework does not modify Standard Model dynamics within experimentally accessible regimes and does not introduce additional fundamental fields. Standard Model phenomenology is recovered as the unsaturated, linear-response regime of the effective description, while finite saturation scales imply necessary high-gradient limits.

The results suggest that mass, charge, and inertia represent complementary limits of a single saturated relational mechanism, providing a unified and falsifiable structural perspective on inertial, gravitational, and electromagnetic response.

1 Introduction

The physical notions of mass, electric charge, and inertia occupy a central role in modern physics. They enter as fundamental parameters in both classical and quantum theories, yet their conceptual origin remains largely unexplained.

In the Standard Model, mass and charge are introduced as intrinsic properties of elementary fields, with numerical values fixed by empirical input. While the Higgs mechanism provides a dynamical account of mass generation for gauge bosons and fermions, it does not address why mass exists as a property, nor why inertial response accompanies it [1]. Electric charge, similarly, is treated as a primitive coupling constant, constrained by gauge symmetry but not derived from a deeper structural principle [2].

From a foundational perspective, inertia presents an additional conceptual challenge. The resistance of a system to acceleration is usually taken as a defining feature of matter, yet its relation to gravitational mass and its possible emergence from more primitive structures have been the subject of long-standing debate [3, 4].

A parallel line of inquiry has emerged in approaches to quantum gravity and pre-geometric physics. In these frameworks, spacetime geometry, and sometimes even locality and causality, are not assumed as fundamental, but are reconstructed as effective descriptions from more primitive relational or algebraic structures [5, 6]. Within such approaches, familiar physical quantities may acquire an emergent or regime-dependent status.

Recent work has emphasized that effective physical descriptions often arise through projections from an underlying configuration space to observable spacetime variables. When such projections are non-injective, multiple underlying configurations correspond to the same effective observable state, leading to intrinsic information loss at the descriptive level [7]. This structural feature has been shown to account for the breakdown of classical probabilistic factorization in quantum correlations, without invoking nonlocal dynamics [8].

In parallel, bounded-response mechanisms have been extensively studied in effective field theories. Born–Infeld electrodynamics provides a paradigmatic example in which divergences are regulated by intrinsic saturation of the field response [9]. Related saturation phenomena appear in gravitational and cosmological contexts, where effective responses cease to scale linearly beyond certain thresholds.

Motivated by these developments, we examine whether mass, electric charge, and inertial response can be understood as manifestations of a single structural mechanism, and whether such an interpretation imposes non-trivial constraints on effective physical descriptions. Specifically, we consider a pre-geometric relational framework in which physical observables arise through a projection subject to intrinsic saturation bounds.

Within this setting, mass is interpreted as an isotropic inhibition of relational relaxation, while electric charge corresponds to an oriented or asymmetric saturation of the same underlying relational flux. Inertia then emerges as the linear-response limit of a bounded projective update process, reflecting the finite resolvability of changes in relational configurations rather than a primitive dynamical property.

The purpose of this work is not to modify the Standard Model or to introduce new fundamental fields. Instead, it is to identify structural constraints on how familiar physical quantities can consistently arise as effective descriptors within a projection-limited

framework. In addition to the conceptual analysis, we provide an explicit effective construction of the proposed saturation mechanism in Lagrangian form, showing that bounded relational projection naturally induces Born–Infeld–type nonlinearities and associated symmetry structures.

The paper is organized as follows. Section 2 introduces the relational and projective framework underlying the analysis. Section 3 identifies the symmetry classes of saturated effective responses and shows how mass, electric charge, and inertial behavior arise as distinct realizations of a bounded-response mechanism. Section 4 examines the consistency of this interpretation with Standard Model phenomenology and identifies necessary high-gradient limits. Section 5 then provides an explicit effective derivation of the saturation mechanism, including the emergence of mass gaps, local $U(1)$ gauge structure, and inertial response as the linear limit of a bounded update process. We conclude in Section 6 with a discussion of conceptual implications, limitations, and possible directions for further investigation.

2 Relational Framework and Projective Description

In this section, we summarize the minimal relational and projective framework required for the present analysis. The detailed construction of this framework is developed in companion works, including the relational and spectral reconstruction of effective geometry [10], the formal role of non-injective projection in effective descriptions [7], and the emergence of bounded-response regimes relevant for mass, charge, and inertia [11]. Here we restrict attention to the structural assumptions that constrain how mass, charge, and inertia can consistently arise as effective descriptors.

2.1 Underlying relational description

We assume that effective spacetime observables arise from an underlying relational description, in which configurations encode admissible correlations between abstract degrees of freedom. No spacetime manifold, metric, or field content is assumed at this level.

Explicit realizations of such relational descriptions, including spectral reconstructions of effective geometry from correlation structure alone, have been developed in companion work [10]. The present paper does not rely on the details of those constructions, but adopts their structural conclusion: spacetime geometry is an effective description rather than a fundamental input.

The underlying relational description should not be interpreted as a hidden-variable completion of existing theories. It defines a space of admissible configurations whose effective description is intrinsically coarse-grained.

Configurations at this level are not endowed with local coordinates, temporal ordering, or intrinsic geometric meaning. These notions arise only when relational configurations admit a stable and approximately injective representation.

2.2 Projection to effective observables

Observable physical quantities are defined through a projection

$$\Pi : \Omega \to \mathcal{O}, \tag{1}$$

where Ω denotes the space of underlying relational configurations and $\mathcal{O}$ the space of effective observables accessible within spacetime descriptions.

A key structural feature of this projection is that it is generally non-injective. Distinct underlying configurations may correspond to the same effective observable state. This identification is not an approximation, but an intrinsic property of the descriptive mapping.

The physical consequences of non-injective projection have been analyzed in detail in the context of quantum correlations [7]. There it was shown that non-injectivity provides a sufficient mechanism for the failure of classical probabilistic factorization, without invoking nonlocal dynamics or hidden-variable assumptions.

In the present work, the same projective structure is assumed to govern inertial and interaction-related quantities. Mass, charge, and inertia are therefore treated as effective descriptors defined on equivalence classes of underlying relational configurations.

2.3 Saturation and bounded response

A central structural requirement of the present framework is that the projection Π admits an intrinsic saturation. Beyond a certain threshold, additional variations in the underlying relational configuration cannot be resolved within the effective description.

This bounded-response behavior has been studied in detail in companion work [11], where it was shown that saturation naturally leads to Born–Infeld-type effective dynamics and to the dynamical selection of stable geometric regimes.

Here, saturation is not introduced as a modification of fundamental dynamics. It is interpreted as a limitation of the effective description itself, reflecting finite projective resolution. As a result, effective responses cannot be extrapolated linearly under arbitrarily large relational gradients.

Operationally, departures from linear behavior are therefore not contingent features, but necessary consequences of finite projective capacity.

2.4 Effective descriptors and regime dependence

Physical quantities such as mass, electric charge, and inertia are treated as effective descriptors defined within the projected space $\mathcal{O}$. They characterize how the effective description responds to variations of the underlying relational configuration.

These quantities are inherently regime dependent. In domains where the projection is approximately injective and unsaturated, classical descriptions are recovered and effective responses behave linearly. In saturated regimes, departures from linearity arise as a direct consequence of projective limitations.

Importantly, the present framework does not modify the operational content of the Standard Model in experimentally tested regimes. It provides a structural reinterpretation of existing quantities while constraining the conditions under which their linear extrapolation remains valid.

Within this perspective, distinctions between mass, charge, and inertial response correspond to different symmetry classes of saturated effective behavior. This classification is developed in the following sections.

3 Symmetry Classes of Saturated Effective Responses

In this section, we identify the symmetry classes of effective physical behavior that arise as stable realizations once saturation of the projective description is taken into account. The goal is not to introduce new physical entities, but to show that mass, electric charge, and inertial response correspond to the only stable symmetry realizations of a bounded effective response, without invoking additional fundamental degrees of freedom.

3.1 Linear and saturated response regimes

When the projection from underlying relational configurations to effective observables is approximately injective, variations at the underlying level are faithfully reflected in the effective description. In this regime, effective responses scale linearly with the magnitude of relational gradients.

Such linear behavior underlies the standard formulation of classical field theories. Small perturbations lead to proportionate responses, and superposition principles apply.

However, when relational gradients exceed the resolving capacity of the effective description, the projection necessarily enters a saturated regime. Beyond this threshold, additional variations at the underlying level no longer produce distinct effective outcomes.

This transition from linear to saturated response is structural rather than dynamical. It reflects a limitation of descriptive resolution, not a modification of underlying laws. Bounded-response regimes of this type are known to give rise to Born–Infeld-like effective dynamics in a variety of contexts [9, 11].

3.2 Isotropic saturation and effective mass

We first consider saturated responses that preserve isotropy in the effective description. In this case, saturation suppresses relational variations uniformly in all directions.

An isotropic inhibition of effective response leads to behavior characteristic of inertial mass and to the symmetry properties underlying gravitational response. The effective description resists changes in motion independently of direction, yielding a scalar parameter that quantifies the degree of saturation.

From this perspective, mass is not introduced as an intrinsic substance. It appears as a measure of how strongly the effective description inhibits relational reconfiguration once saturation is reached.

This interpretation is consistent with the equivalence between inertial and gravitational mass. Both correspond to isotropic limitations of effective response, differing

only in the context in which the response is probed. The universality of free fall follows as a direct consequence of the symmetry of the saturation mechanism.

3.3 Oriented saturation and effective charge

We now consider saturated responses that break isotropy while preserving locality and stability of the effective description. In this class, saturation is directionally biased with respect to relational gradients.

Such oriented saturation necessarily leads to behavior characteristic of electric charge. The effective response distinguishes between opposing directions, giving rise to attractive and repulsive interactions depending on orientation.

Charge thus appears as a signed quantity associated with asymmetric saturation of relational flux. The existence of both positive and negative charges reflects the presence of two stable orientations of the saturated response.

Importantly, this interpretation does not require introducing charge as a primitive coupling. It arises as a symmetry-breaking mode of the same bounded-response mechanism that gives rise to mass.

The long-range character of electromagnetic interactions follows from the fact that oriented saturation does not induce isotropic suppression, allowing extended field-like behavior within the effective description.

3.4 Inertia as finite-resolution response

Inertial behavior emerges when changes in motion probe the finite update capacity of the effective description. Acceleration corresponds to a demand for rapid reconfiguration of relational correlations within the projected space.

When such reconfiguration remains well below the projective capacity, the effective response is linear. In this regime, resistance to acceleration scales proportionally with the applied stress, reproducing the standard inertial relation as a linear-response approximation.

As the required rate of reconfiguration approaches the resolving limit of the projection, the effective response saturates. Beyond this point, additional relational variation cannot be resolved instantaneously, and the projected dynamics exhibits a bounded update behavior.

In this view, inertia is not a fundamental property attached to matter. It is the linear-response limit of a bounded projective update process, reflecting the finite resolvability of variations in relational configurations within the effective description.

This interpretation preserves all empirical content of relativistic kinematics while reinterpreting inertial resistance as a consequence of finite descriptive capacity rather than a primitive dynamical postulate.

3.5 Unified classification

The analysis above yields a classification of the *minimal* effective physical responses permitted by a bounded projective description under a single scalar saturation constraint.

Within this scope, three stable response classes are singled out. Isotropic saturated responses correspond to mass-like behavior. Oriented saturated responses correspond to charge-like behavior. Finite-resolution resistance to reconfiguration corresponds to inertia.

All three arise from the same structural mechanism: saturation of the effective description under non-injective projection. Their distinction follows from symmetry properties of the saturation manifold rather than from distinct underlying substances or independent fundamental fields.

This classification establishes existence rather than completeness. It shows that, once bounded projective capacity is assumed, mass-like, charge-like, and inertial responses necessarily appear as the minimal symmetry classes compatible with a scalar saturation bound. It does not yet claim to exhaust all possible gauge structures.

In particular, partially anisotropic or higher-rank saturation patterns are not excluded. Such patterns correspond to saturation manifolds with higher-dimensional internal degeneracies. For an N-component relational multiplet subject to a scalar bound $\rho^2 = \vec{\phi} \cdot \vec{\phi}$, the symmetry group preserving saturation is $O(N)$, whose connected component $SO(N)$ becomes non-abelian for $N \geq 3$. Local compatibility of such degeneracies naturally leads to non-abelian gauge redundancies.

Accordingly, the present work identifies the minimal rank-one realization of bounded-response symmetry classes and delineates a concrete extension path toward higher-rank oriented saturation. The embedding of the full Standard Model gauge group is deferred to a dedicated analysis of multi-channel projection and stability conditions.

In the following section, we examine the consistency of this framework with established phenomenology and show that bounded-response symmetry classes imply necessary limits on linear extrapolations.

4 Consistency with Standard Model Phenomenology

In this section, we examine the compatibility of the proposed structural interpretation with established phenomenology of the Standard Model. The purpose is not to derive Standard Model parameters, but to ensure that the framework does not conflict with known experimental facts or operational definitions. We further identify necessary physical consequences that must follow if the saturated-response interpretation is meaningful.

4.1 Status of mass in the Standard Model

In the Standard Model, particle masses arise through the Higgs mechanism, which endows fermions and gauge bosons with effective mass terms via spontaneous symmetry breaking. This mechanism successfully accounts for the dynamical generation of mass and its role in particle interactions.

The present framework does not challenge this description. Instead, it addresses a logically distinct question: why mass appears as a universal scalar measure of inertial response and of the coupling to gravitational interaction.

Within the proposed interpretation, the Higgs mechanism determines how mass values are assigned within the effective description, while the structural origin of mass

as an isotropic saturated response explains why such a parameter has the physical meaning it does. The two perspectives operate at different conceptual levels and are therefore complementary rather than competing.

4.2 Electric charge and gauge structure

Electric charge in the Standard Model is associated with local gauge invariance and conserved currents. Its quantization and coupling structure are fixed by the underlying gauge symmetry and anomaly cancellation requirements.

The framework developed here does not modify gauge symmetry or charge conservation. It reinterprets electric charge as an effective manifestation of oriented saturation in the projective response, without altering its operational role.

From this perspective, gauge invariance constrains how oriented saturation can appear consistently within the effective description. The existence of discrete charge values reflects the stability of specific oriented saturation modes, rather than an arbitrary assignment of coupling constants.

Importantly, this interpretation does not predict deviations from known electromagnetic phenomena in the weak-field regime. All standard results of quantum electrodynamics are recovered in the unsaturated and approximately injective limit.

4.3 Inertia, relativistic dynamics, and equivalence

Relativistic dynamics treats inertia as a fundamental response encoded in the energy–momentum relation. The equivalence between inertial and gravitational mass is experimentally well established and constitutes a cornerstone of relativistic physics.

Within the present framework, this equivalence arises naturally from the isotropy of saturated response. Both inertial resistance to acceleration and gravitational response correspond to the same structural limitation of effective reconfiguration. Inertia itself is shown to emerge as the linear-response limit of a bounded projective update process, while gravitational response probes the same saturation mechanism through spacetime geometry.

No modification of relativistic kinematics is implied in the unsaturated regime. Lorentz invariance remains an effective symmetry of the projected description as long as the projection is approximately injective. Lorentz invariance remains an effective symmetry of the projected description, valid in regimes where the projection is approximately injective and unsaturated.

4.4 Absence of observable deviations at accessible scales

A crucial consistency requirement is the absence of observable deviations from Standard Model predictions in experimentally tested regimes. The framework satisfies this requirement by construction.

Saturation effects become relevant only when relational gradients approach the resolving capacity of the effective description. In present-day particle physics experiments, electromagnetic and inertial responses remain well within the linear regime, and no saturation is probed.

As a result, the framework predicts no departures from established cross sections, decay rates, or precision tests of quantum electrodynamics and electroweak theory. All Standard Model phenomenology is recovered as the linear-response limit of the effective description.

4.5 High-energy consistency and saturation limits

The absence of deviations at accessible scales does not imply that effective responses can be extrapolated linearly to arbitrarily high gradients. If mass, charge, and inertia arise as saturated responses constrained by a finite projective bound b, then strictly unbounded growth of effective couplings would render the framework internally inconsistent.

The consistency of the saturated-response interpretation therefore requires the existence of regimes in which perturbative extrapolations must fail. Such deviations are not additional assumptions or phenomenological predictions, but necessary consequences of introducing a finite projective capacity.

An empirical indication that such bounded behavior already exists is provided by the well-established Schwinger limit, which sets an upper threshold on sustainable electromagnetic field invariants before vacuum instability occurs. In the present framework, this threshold is interpreted as the operational manifestation of a finite projective bandwidth of the vacuum.

We emphasize that the present work does not address the microscopic dynamics of vacuum instability or pair production. The detailed realization of saturation effects in strong-field quantum electrodynamics will be developed in a companion paper. Here, the Schwinger limit serves solely as an existence proof that unbounded extrapolation of effective responses is physically untenable.

This interpretation is explicitly falsifiable. If effective couplings can be shown to grow without bound under arbitrarily strong field gradients, with no indication of saturation or inflection, then the bounded-response framework developed here must be rejected.

4.6 Relation to other bounded-response frameworks

Bounded-response mechanisms have previously been considered in both electromagnetic and gravitational contexts. Born–Infeld electrodynamics provides a well-known example in which saturation regulates divergent field strengths without spoiling low-energy phenomenology [9].

The present framework generalizes this idea conceptually. Rather than introducing bounded response as a modification of specific field equations, saturation is interpreted as a generic limitation of effective descriptions arising from non-injective projection.

This shift in perspective allows mass, charge, and inertia to be treated on equal footing, as different symmetry realizations of the same structural mechanism.

4.7 Summary

The proposed interpretation is fully consistent with Standard Model phenomenology. It does not alter gauge structure, particle content, or dynamical equations in experimentally tested regimes.

At the same time, it identifies necessary high-gradient limits in which linear extrapolations of effective responses must break down. Standard Model physics is recovered as the unsaturated regime of a bounded projective description.

In the following section, we discuss conceptual implications, limitations, and possible directions for further investigation.

5 Formal Derivation of Saturated Responses

This section adds the minimal mathematical structure required to connect the relational framework to effective physical response. It provides an explicit effective action for the projected relaxation degree of freedom and shows how mass-like, charge-like, and inertial responses arise under finite projective capacity. The goal is not to postulate new fundamental fields, but to identify the minimal effective structures necessarily induced by bounded relational update flux.

Scope of Section 5.

This section provides a formal effective derivation of mass-like, charge-like, and inertial responses from bounded projective capacity. The objective is to establish existence and internal consistency rather than microscopic uniqueness. Completeness with respect to the full Standard Model gauge structure is not claimed and is deferred to higher-rank oriented saturation analyses. Throughout this section, the term "derivation" is used in this effective sense, with all closure assumptions stated explicitly.

5.1 Relational flux and projective capacity

We represent the local relational update content by a rank-two tensor density

$$J_{\mu\nu} \equiv \partial_\mu \phi \, \partial_\nu \phi, \tag{2}$$

where ϕ encodes the effective relaxation mode surviving projection. The projection Π is assumed to have finite capacity. Operationally, this means that the observable description cannot resolve arbitrarily large values of relational flux invariants constructed from $J_{\mu\nu}$. We parametrize this limitation by introducing a saturation scale b with dimensions of a flux density.

The minimal Lorentz-invariant completion enforcing boundedness while recovering the standard quadratic kinetic term at low flux is the Dirac–Born–Infeld-type Lagrangian

$$\mathcal{L}_{\mathrm{BI}}(\phi) \equiv b^2 \left(1 - \sqrt{1 - \frac{\partial_\mu \phi \, \partial^\mu \phi}{b^2}} \right). \tag{3}$$

Equivalently, the associated action reads

$$S_\Pi[\phi] = -b^2 \int d^4x \left(\sqrt{-\det\left(\eta_{\mu\nu} - b^{-2} \partial_\mu \phi \, \partial_\nu \phi\right)} - 1 \right), \tag{4}$$

making explicit that saturation corresponds to an effective deformation of the metric volume element as relational flux approaches b.

For weak gradients, $\partial\phi \ll b$, Eq. (3) expands as

$$\mathcal{L}_{\rm BI}(\phi) = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi + \frac{1}{8b^2}\left(\partial_\mu\phi\,\partial^\mu\phi\right)^2 + \mathcal{O}\big(b^{-4}\big)\,, \tag{5}$$

recovering the linear-response regime.

5.2 Isotropic saturation and the emergence of a mass scale

Isotropic saturation corresponds to saturation reached through the scalar invariant

$$X \equiv \partial_\mu\phi\,\partial^\mu\phi, \tag{6}$$

rather than through direction-dependent components.

Equation (3) enforces boundedness but does not generate a mass gap. A gap arises when isotropic saturation selects a typical finite-flux background. In a projection-limited regime, X aggregates many unresolved relational configurations. Under a fixed capacity bound, coarse-graining produces concentration of measure around a typical saturation value $X_\star$.

We encode this statistical rigidity through the effective completion

$$\mathcal{L}_{\rm iso} \equiv \mathcal{L}_{\rm BI}(\phi) - \frac{\kappa}{2}\left(X - X_\star\right)^2\,, \tag{7}$$

where $X_\star$ denotes the saturation-typical value. We stress that the quadratic form is not claimed to be uniquely derived. The existence of a typical value follows from concentration arguments, but the detailed functional form of the penalty reflects a minimal local and analytic closure. Alternative closures are not excluded. The role of Eq. (7) is to demonstrate the existence of a mass gap under isotropic saturation, not to provide a microscopic derivation of the penalty.

Let $\phi = \phi_0 + \delta\phi$ with $X[\phi_0] = X_\star$. Expanding to second order yields

$$\mathcal{L}_{\delta\phi} \simeq \frac{1}{2}Z(X_\star, b)\,\partial_\mu\delta\phi\,\partial^\mu\delta\phi - \frac{1}{2}\kappa\,(\delta\phi)^2, \tag{8}$$

with

$$Z(X_\star, b) \equiv \left(1 - \frac{X_\star}{b^2}\right)^{-3/2}. \tag{9}$$

The equation of motion takes the Klein–Gordon form

$$\left(\Box + m_{\rm eff}^2\right)\delta\phi = 0, \tag{10}$$

where

$$m_{\rm eff}^2 = \frac{\kappa}{Z(X_\star, b)}. \tag{11}$$

5.3 Inertia as the linear limit of saturated update rates

To relate bounded DBI kinetics to inertial response, we consider a particle-like reduction. Specifically, we focus on homogeneous or slowly varying modes for which the dominant effect of saturation is captured by a single collective coordinate $x(t)$. This reduction is valid when spatial gradients are subdominant and the observable probes global reconfiguration rates rather than local field structure.

For such a mode, the DBI kinetic sector induces the generalized momentum

$$p = \frac{m_{\text{eff}}\,\dot{x}}{\sqrt{1-\dot{x}^2/a_\star^2}}, \tag{12}$$

where $a_\star$ denotes the maximal admissible update rate. Defining $\mathcal{F} \equiv dp/dt$, the equation of motion becomes

$$\ddot{x} = \frac{\mathcal{F}}{m_{\text{eff}}}\left[1+\left(\frac{\mathcal{F}}{m_{\text{eff}}a_\star}\right)^2\right]^{-1/2}. \tag{13}$$

This equation follows from the Legendre transform of the DBI kinetic term and exhibits a linear regime together with asymptotic saturation.

The phenomenological form

$$\frac{d^2\mathcal{O}}{dt^2} = a_\star\,\mathcal{S}\left(\frac{\mathcal{F}}{m_{\text{eff}}a_\star}\right), \tag{14}$$

with $\mathcal{S}(x) = \tanh(x)$, represents a smooth analytic closure of the same universality class. Inertia thus emerges as the linear-response limit of bounded update dynamics.

5.4 Oriented saturation and the emergence of a local $U(1)$ symmetry

If at least two relational degrees of freedom remain unresolved, the saturation-preserving manifold is necessarily a compact connected one-dimensional space, uniquely S^1.

This residual S^1 structure can be interpreted geometrically as the phase fiber associated with a non-injective projection of the underlying relational configuration space. When multiple microscopic configurations correspond to the same effective state, the effective description admits a compact fiber representing the remaining undetermined phase. The resulting phase degree of freedom naturally carries a $U(1)$ symmetry, and oriented saturation of the relaxation flux defines a conserved invariant associated with this fiber.

A minimal realization is

$$\psi \equiv \rho\, e^{i\theta}, \tag{15}$$

with ρ fixed near $\rho_\star$ and θ underdetermined. The redundancy

$$\theta(x) \to \theta(x) + \alpha(x) \tag{16}$$

requires a compensating connection. Introducing

$$D_\mu \psi \equiv (\partial_\mu - iqA_\mu)\psi, \tag{17}$$

yields

$$\mathcal{L}_{\text{ori}} = b^2 \left(1 - \sqrt{1 - \frac{D_\mu \psi \, D^\mu \psi^*}{b^2}}\right) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \tag{18}$$

5.5 Immediate falsifiability

The effective action (4) ceases to define a real description when

$$\partial_\mu \phi \, \partial^\mu \phi > b^2. \tag{19}$$

The framework is ruled out if strictly linear behavior is experimentally required beyond this scale.

Universality of b provides a second falsifiability channel. Identifying the saturation threshold with the Schwinger field $E_{\text{S}} \sim b/e$ yields the benchmark

$$b \sim 10^{24} \, \text{J} \, \text{m}^{-3}, \tag{20}$$

up to order-unity factors. Failure of cross-sector consistency between inertial, mass, and strong-field phenomena would falsify the bounded-response hypothesis.

6 Discussion, Limitations, and Outlook

In this final section, we summarize the conceptual implications of the proposed framework, clarify its limitations, and outline possible directions for future investigation. While the analysis remains primarily structural, it now includes an explicit effective realization of the saturation mechanism. This allows physical constraints to be formulated at the level of an effective Lagrangian, rather than at the level of interpretation alone.

6.1 Conceptual implications

A central implication of the present analysis is that mass, electric charge, and inertia need not be treated as fundamentally distinct physical primitives. Instead, they can be consistently understood as different symmetry realizations of a single structural mechanism, namely saturation of effective response under non-injective projection.

From this perspective, the apparent diversity of physical properties reflects differences in how the effective description responds to relational variation, rather than differences in underlying substance. Scalar, oriented, and finite-bandwidth response modes correspond respectively to mass-like, charge-like, and inertial behavior, with inertia arising as the linear-response limit of a bounded projective update process.

The explicit effective construction introduced in Section 5 shows that these distinctions can be realized within a single bounded-response framework, without modifying the operational definitions of mass and charge. In this sense, the analysis establishes

existence and internal consistency of these response classes rather than a unique microscopic derivation. Mass appears as a spectral gap induced by isotropic saturation, while electric charge emerges as the generator of a local phase degeneracy associated with oriented saturation. Geometrically, this degeneracy corresponds to the phase fiber induced by non-injective projection of the underlying relational configuration space, which naturally carries a $U(1)$ symmetry. Inertia, in turn, corresponds to the first-order expansion of a saturated update dynamics.

This interpretation provides a unified conceptual basis for the equivalence between inertial and gravitational mass and for the signed nature of electric charge, while preserving their empirical roles within established physical theories.

6.2 Relation to existing foundational approaches

The framework developed here is compatible with a wide range of foundational approaches in which spacetime and physical observables are regarded as effective constructs. It does not rely on a specific microscopic ontology and can be embedded in different relational, background-independent, or pre-geometric settings.

Unlike approaches that postulate new degrees of freedom or modified fundamental dynamics, the present work operates deliberately at the level of effective description. Its contribution is to isolate a minimal structural mechanism that constrains how familiar physical notions can consistently arise within effective theories subject to finite descriptive capacity. Within this perspective, gauge structures need not be introduced as independent symmetry postulates. Instead they can arise from degeneracies of the effective description induced by non-injective projection, with phase symmetries appearing as residual fiber degrees of freedom of the projection.

In this sense, the framework complements rather than replaces existing formulations. It provides an interpretative and constraining layer that may coexist with standard quantum field theory and relativistic dynamics, while clarifying the domain of validity of linear-response extrapolations.

6.3 Limitations of the present work

The present analysis is subject to several important limitations.

First, no microscopic dynamics of the underlying relational description is specified. As a result, the framework does not claim to derive unique functional forms for all effective closures, nor to predict numerical values for particle masses, charges, or coupling constants. The present analysis identifies structural invariants associated with these quantities but leaves their numerical values to be fixed by the detailed microscopic realization of the relational dynamics. These remain empirical inputs determined within effective theories operating in the unsaturated regime.

Second, the specific form of the effective penalty associated with isotropic saturation is introduced as a minimal analytic closure. While the existence of a typical saturation value follows from concentration arguments, the detailed functional form of the penalty is not uniquely fixed at this level of description and is deferred to future work.

Third, the analysis does not provide detailed quantitative predictions for high-energy or strong-field deviations from Standard Model behavior. Nevertheless, the

effective formalism developed in Section 5 identifies a well-defined departure from strictly linear inertial response and introduces characteristic saturation scales, implying necessary limits on linear extrapolations.

Finally, the framework does not address particle generations, flavor structure, or symmetry-breaking patterns within the Standard Model. Its scope is restricted to the structural origin and effective interpretation of mass, charge, and inertia.

6.4 Possible extensions and outlook

Despite these limitations, the framework suggests several concrete directions for further investigation.

One natural extension is the construction of explicit relational or spectral models in which the projective bound and its associated saturation scale can be derived rather than postulated. Such developments would allow the present effective closures to be replaced by microscopically grounded expressions.

Another important direction concerns the systematic study of strong-field and high-gradient regimes, where saturation effects are expected to become operational. Clarifying the relation between bounded-response mechanisms and known nonlinear phenomena in quantum field theory may help identify experimentally accessible signatures.

More broadly, the analysis invites reconsideration of the status of physical parameters traditionally regarded as fundamental. If mass and charge are effective descriptors tied to the finite descriptive capacity of a non-injective effective projection, their role in physical theories may be understood as constrained rather than primitive.

6.5 Concluding remarks

We have proposed a unified structural interpretation of mass, electric charge, and inertia as effective manifestations of saturated response under non-injective projection. The framework is fully consistent with Standard Model phenomenology and does not modify its dynamical content in experimentally tested regimes.

By providing an explicit effective realization of the saturation mechanism, the present work goes beyond purely interpretative unification and introduces well-defined structural constraints. In particular, the emergence of inertia as the linear limit of a bounded update process implies necessary high-gradient limits, rendering the framework falsifiable in principle.

While deliberately modest in its quantitative claims, the analysis establishes a coherent and internally consistent basis for further exploration of the structural origin of physical properties.

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